

Ninth Section.

Estimation of the Roots.

§ 99. The Budan–Fourier Theorem

The functions of Sturm chains do solve, completely and without exception, the problem of determining the number of roots of an equation between prescribed limits; but their actual computation is usually difficult and often impracticable. There is a series of theorems for estimating the number of roots between given limits. They supply much simpler rules, but of course they do not answer the question completely: they give only a maximum beyond which the number of roots cannot go. These rules are often very useful and sufficient in applications, and therefore they must not be omitted here.

We first consider a procedure due to Budan and afterwards used and extended by Fourier.¹ In place of a Sturm chain let us consider the sequence of derivatives of a real function $f(x)$ of degree n ,

$$f(x), \quad f'(x), \quad f''(x), \dots, \quad f^{(n)}(x), \quad (1)$$

where $f^{(n)}(x)$ is a constant, which we suppose different from zero and positive. Apart from a positive numerical factor it is the coefficient of x^n in $f(x)$.

Let α and $\beta > \alpha$ again be two real numbers determining the interval (α, β) in which the real roots of $f(x)$ are to be counted. We shall first suppose that, in the interval (α, β) , no two members of the sequence (1) vanish at the same time, and that for $x = \alpha$ and $x = \beta$ no member of the sequence vanishes.

Let x increase continuously from α to β . A change in the succession of signs of (1) can occur only when one of the functions passes through zero. If $f^{(\nu)}(x)$ passes through zero for $x = \xi$, then, according as $f^{(\nu+1)}(\xi)$ is positive or negative, $f^{(\nu)}(x)$ passes from negative to positive values or from positive to negative values; hence between $f^{(\nu)}$ and $f^{(\nu+1)}$ one variation of sign is lost in passing through ξ . This also applies when $f^{(\nu)}$ is the original function $f(x)$ itself. If, however, $f^{(\nu)}$ is one of the derivatives, then a function $f^{(\nu-1)}$ precedes it. Since $f^{(\nu-1)}$ is not zero at $x = \xi$ by hypothesis, while $f^{(\nu)}$ changes its sign in passing through ξ , between $f^{(\nu-1)}$ and $f^{(\nu)}$ either one variation is lost or one variation is gained. Thus, if an interior member of the sequence (1) passes through zero, the number of variations remains unchanged or two variations are lost. If $f(x)$ itself passes through zero, one variation is lost.

This gives the theorem:

I. The number of roots of $f(x)$ lying between α and β is at most equal to the number of variations lost between α and β ; and, if it is smaller, the difference is an even number.

In formula form this may be stated as follows. If $V(x)$ denotes the number of sign variations displayed by the sequence (1) for a value x , then the number of roots lying between α and β is

$$V(\alpha) - V(\beta) - 2h, \quad (2)$$

where h is a non-negative integer.

The proof of I still requires a supplement for the case in which several consecutive members of (1) vanish simultaneously. Suppose that, for a value ξ of x between α and β , all the members in the sequence

$$f^{(\nu)}(x), \quad f^{(\nu+1)}(x), \dots, f^{(\nu+\mu-1)}(x), \quad f^{(\nu+\mu)}(x) \quad (3)$$

¹Budan, *Nouvelle méthode pour la résolution des équations numériques*, submitted to the Paris Academy in 1803. Fourier, *Analyse des équations déterminées*. Paris 1831. For the history of this question compare Lagrange, *Traité de la résolution des équations*, Note VIII (Works, vol. 8).

except the last vanish, and let the last, $f^{(\nu+\mu)}(\xi)$, be positive. We mark off around ξ two intervals δ_1, δ_2 , with all values in δ_1 smaller than ξ and all values in δ_2 larger than ξ , and choose them so small that the functions (3) vanish in them only at ξ ; thus $f^{(\nu+\mu)}(x)$ remains positive. Since, when $f^{(\nu+\mu)}(x)$ is positive, $f^{(\nu+\mu-1)}(x)$ increases with x , the latter is negative in δ_1 and positive in δ_2 . It follows that $f^{(\nu+\mu-2)}(x)$ decreases in δ_1 and increases in δ_2 , and is therefore positive in both intervals. Continuing in this way, one obtains the sign sequence

$$\begin{array}{c|ccccc} & f^{(\nu)} & f^{(\nu+1)} & f^{(\nu+2)} & \dots & f^{(\nu+\mu)} \\ \hline \delta_1 & (-1)^\mu & (-1)^{\mu-1} & (-1)^{\mu-2} & \dots & + \\ \delta_2 & + & + & + & \dots & + \end{array} \quad (4)$$

for the sequence (3). Thus in passing through ξ a row consisting entirely of variations is transformed into a row of permanences; μ variations are lost in the sequence (3). If $f^{(\nu+\mu)}(\xi)$ is negative, all the signs in (4) are reversed, and the conclusion is otherwise the same.

If now $\nu = 0$, so that $f^{(\nu)}(x)$ is the original function $f(x)$, then f has at ξ a root of multiplicity $\mu + 1$, and in the sequence (1) a loss of μ variations also occurs. If, however, $\nu > 0$ and the function preceding $f^{(\nu)}(x)$, namely $f^{(\nu-1)}(x)$, is not zero, then the following is obtained: for even μ no variation is lost between $f^{(\nu-1)}$ and $f^{(\nu)}$; for odd μ one variation is lost or one variation is gained. Hence the loss of variations in passing through ξ in the sequence

$$f^{(\nu-1)}, \quad f^{(\nu)}, \quad f^{(\nu+1)}, \dots, f^{(\nu+\mu)} \quad (5)$$

is equal to μ for even μ , and to $\mu + 1$ or $\mu - 1$ for odd μ ; it is therefore always an even number and never negative.

This result can be expressed analytically in a form that makes its meaning clearer. Let $V(x)$ denote, as before, the number of variations in (1) for a specified value of x , with the additional convention that, if some members vanish, they are simply omitted in counting variations. Then $V(x)$ may be treated as a function of x , though it changes only by integral jumps and is discontinuous at the values of x for which some members of (1) vanish. Let $V(x-0)$ and $V(x+0)$ denote the values immediately before and immediately after such a point. The preceding discussion gives, in all cases,

$$V(\xi+0) = V(\xi), \quad (6)$$

and, if ξ is a root of $f(x)$ of multiplicity μ ,

$$V(\xi-0) = V(\xi) + \mu + 2h, \quad (7)$$

where h is a non-negative integer.

It follows, as long as none of the functions (1) vanishes at $x = \alpha$ or $x = \beta$, that $V(\alpha) - V(\beta)$ differs from the sum of the multiplicities μ , that is from the number of roots in (α, β) counted with multiplicity, by an even, non-negative number. Thus Theorem I is freed from its earlier restriction.

We may also drop the assumption that none of the functions (1) vanishes at α and β , provided that α and β are not roots of $f(x)$. The theorem may then be applied to two values, the first slightly greater than α , the second slightly less than β , enclosing the same roots. If λ is the number of these roots, then

$$V(\alpha+0) - V(\beta-0) = \lambda + 2h.$$

Since

$$V(\alpha+0) = V(\alpha), \quad V(\beta-0) = V(\beta) + 2h',$$

it follows that

$$V(\alpha) - V(\beta) = \lambda + 2(h + h'),$$

again the expression of our theorem. If, however, α or β itself is a root of $f(x)$, then the theorem is correct only when these limiting values are not counted as belonging to the interval.

The Budan–Fourier theorem does not, like Sturm’s theorem, give a completely certain decision about the number of roots in an interval. Yet in many cases it can replace Sturm’s theorem. If the interval (α, β) has been narrowed so far that no variation, or only one variation, is lost from α to β , then it follows with certainty that in the first case no root, and in the second case exactly one root, lies in the interval. The separation of the roots can therefore be given completely by Budan’s sequence only when several variations are not lost simultaneously in passing through a value ξ . This behaviour cannot be recognized with certainty, however small the interval (α, β) is chosen; consequently, if after a suitable narrowing of the interval the separation has not succeeded, one must pass to another procedure, in the last resort to Sturm’s theorem.

§ 100. Newton’s Rule

A supplement to the preceding rule, one that gives a closer approximation to the exact fixing of root bounds, is the rule now to be discussed, due to Newton but first proved by Sylvester.²

Let again

$$f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0 \quad (1)$$

be the equation to be examined. With $n(m)$ denoting, as usual, the product $1 \cdot 2 \cdot 3 \cdots m$, and with $f^{(\nu)}(x)$ denoting the ν th derivative, put

$$f_\nu(x) = \frac{(n - \nu)!}{n!} f^{(\nu)}(x), \quad (2)$$

$$F_\nu(x) = f_\nu(x)^2 - f_{\nu-1}(x)f_{\nu+1}(x), \quad (3)$$

with the addition that

$$F_0 = F_n = 1, \quad F_n = f_n^2, \quad (4)$$

so that these end terms, which is essential, are positive. From (2) and (3) one obtains, for the derivatives,

$$f'_\nu(x) = (n - \nu)f_{\nu+1}(x), \quad (5)$$

$$F'_\nu(x) = (n - \nu + 1)(f_{\nu-1}F_{\nu+1} + F_{\nu-1}f_{\nu+1}), \quad (6)$$

up to positive numerical factors in the only sense relevant for signs. Since positive factors do not matter, Budan’s sequence may be replaced by

$$f_0, \quad f_1, \quad f_2, \dots, \quad f_n.$$

Newton’s rule now uses the double row

$$\begin{array}{cccccc} f_0, & f_1, & f_2, & \dots, & f_{n-1}, & f_n, \\ F_0, & F_1, & F_2, & \dots, & F_{n-1}, & F_n. \end{array} \quad (7)$$

For two consecutive columns, say

$$\begin{array}{cc} f_\nu, & f_{\nu+1} \\ F_\nu, & F_{\nu+1}, \end{array}$$

and for a value of x at which none of the four functions vanishes, the following possibilities for the

²Newton, *Arithmetica universalis*. Sylvester, *Transactions of the Royal Irish Academy*, vol. 24 (1871); *Philosophical Magazine*, fourth series, vol. 31. Compare also Petersen, *Theorie der algebraischen Gleichungen*. Copenhagen 1878.

succession of signs may occur:

a)	++	++	--	--	PP
	++	--	++	--	
b)	+-	+-	-+	-+	VV
	+-	-+	+-	-+	
c)	++	++	--	--	PV
	+-	-+	+-	-+	
d)	+-	+-	-+	-+	VP
	++	--	++	--	

We designate these behaviours as follows:

- a) permanence–permanence PP ,
- b) variation–variation VV ,
- c) permanence–variation PV ,
- d) variation–permanence VP .

The two theorems to be proved are the following.

II. The number of roots of $f(x)$ lying between α and β is either exactly equal to, or smaller by an even number than, the number of variation–permanences lost in the double row (7) in passing from α to β .

III. The number of roots of $f(x)$ lying between α and β is either exactly equal to, or smaller by an even number than, the number of permanence–permanences gained in the double row (7) in passing from α to β .

Since these two theorems do not always give the same upper bound for the number of roots, one will use whichever gives the lower bound.

In the proof we make the following hypotheses, from some of which we shall later free ourselves.

1. $f(x)$ has no double or multiple roots.
2. Of the functions $f_\nu(x)$ no two adjacent ones vanish for the same value of x .
3. Of the functions $F_\nu(x)$ no two adjacent ones vanish for the same value of x .
4. A consequence of 2, by formula (3), is that $f_\nu(x)$ and $F_\nu(x)$ do not vanish for the same value of x .
5. At $x = \alpha$ and $x = \beta$ none of the functions f_ν, F_ν vanishes.

The proof of II and III is then as follows. Let x increase continuously from α to β . A change in the sign sequence can occur only when one of the functions f_ν or F_ν passes through zero; it is therefore enough to examine the effect of such a passage.

Suppose first that one of the middle functions f_ν passes through zero, so that $0 < \nu < n$. According as $f_{\nu+1}$ is positive or negative, f_ν increases or decreases. Thus in the first case $f_\nu(x-0)$ is negative and $f_\nu(x+0)$ positive, and in the second case the reverse occurs. Further, (3) shows that, if $f_\nu = 0$, then $F_{\nu-1}$ and $F_{\nu+1}$ are positive, while F_ν has the sign opposite to that of $f_{\nu-1}f_{\nu+1}$. The remaining possibilities are therefore

	$\nu-1$	ν	$\nu+1$		$\nu-1$	ν	$\nu+1$
$f(x-0)$	+	–	+	$f(x-0)$	–	–	+
$f(x+0)$	+	+	+	$f(x+0)$	–	+	+
F	+	–	+	F	+	+	+

and hence

$$\begin{array}{c|c} x-0 & VV, VV \\ x+0 & PV, PV \end{array} \quad \begin{array}{c} PP, VP \\ VP, PP. \end{array}$$

Thus neither a VP nor a PP is lost or gained.

If $f(x)$ itself passes through zero, then, taking $f_1(x)$ positive, which is enough in view of the preceding remark, we have

$$\begin{array}{c|c} & 0 \quad 1 \\ \hline f(x-0) & - \quad + \\ f(x+0) & + \quad + \\ \hline F & + \quad + \end{array}$$

so that a VP changes into a PP : one VP is lost and one PP is gained.

Finally let F_ν pass through zero, necessarily with $0 < \nu < n$. If $F_\nu = 0$, then $f_{\nu-1}$ and $f_{\nu+1}$ must have the same sign; otherwise F_ν would consist of two positive terms and could not vanish. Moreover, by (6) the derivative $F'_\nu(x)$ has the same sign as $f_{\nu-1}f_\nu F_{\nu+1}$. Hence $F_\nu(x-0)$ has the opposite sign and $F_\nu(x+0)$ the same sign as that product. Omitting cases arising merely by simultaneous reversal of signs, one obtains

$$\begin{array}{c|ccc} & \nu-1 & \nu & \nu+1 \\ \hline f & + & + & + \\ F(x-0) & + & - & + \\ F(x+0) & + & + & + \end{array} \quad \begin{array}{c|ccc} & \nu-1 & \nu & \nu+1 \\ \hline f & + & + & + \\ F(x-0) & - & - & + \\ F(x+0) & - & + & + \end{array}$$

$$\begin{array}{c|ccc} & \nu-1 & \nu & \nu+1 \\ \hline f & + & - & + \\ F(x-0) & + & + & + \\ F(x+0) & + & - & + \end{array} \quad \begin{array}{c|ccc} & \nu-1 & \nu & \nu+1 \\ \hline f & + & - & + \\ F(x-0) & - & + & + \\ F(x+0) & - & - & + \end{array}$$

and the variations in the four cases are

$$\begin{array}{c|c} x-0 & PV, PV \quad PP, PV \quad VP, VP \quad VV, VP \\ x+0 & PP, PP \quad PV, PP \quad VV, VV \quad VP, VV. \end{array}$$

In the first case two PP 's are gained; in the second and fourth no change occurs; in the third two VP 's are lost. Theorems II and III are thereby proved.

The assumptions 2 and 3, that no two adjacent functions f_ν and no two adjacent functions F_ν vanish for the same x , may be removed. If one retains the first assumption, that there are no multiple roots, then one can change the coefficients a_ν by arbitrarily small amounts so that 2 and 3 are satisfied and, at the same time, the roots between α and β move so little that none leaves the interval. Since the roots occur singly, real roots cannot thereby pass into imaginary ones. Whether the theorem, with proper counting, remains valid for multiple roots as well may be left open here.

§ 101. Descartes' Rule

Descartes' rule, also named after Harriot,³ gives a rule for estimating the number of positive roots. We can regard it as a special case of Budan's theorem by taking $\alpha = 0$ and $\beta = \infty$, or at least so large that

$$f(x), \quad f'(x), \quad f''(x), \dots, \quad f^{(n)}(x) \quad (1)$$

all have the same sign for $x > \beta$. This sign is the sign of the coefficient of the highest power of x in $f(x)$, and may be assumed positive.

³Harriot, *Artis analyticae praxis ad aequationes algebraicas resolvendas*. 1631. Gauss' Werke, vol. III, p. 67. Lagrange, l. c.

If

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n,$$

then, apart from positive numerical factors, the functions (1) take at $x = 0$ the values

$$a_n, \quad a_{n-1}, \quad a_{n-2}, \dots, \quad a_0. \quad (2)$$

This gives the theorem:

IV. The number of positive roots of $f(x) = 0$ is equal to, or smaller by an even number than, the number of sign variations in the sequence of coefficients of $f(x)$, zero coefficients being simply omitted and the root $x = 0$, if present, not being counted.

For negative roots one may either apply Budan–Fourier to the interval $(-\infty, 0)$, or replace x by $-x$, that is, change the signs of $a_1, a_3, a_5, \dots$, and then apply IV.

In the same manner we specialize Newton’s criterion of § 100. We observe that

$$f_\nu(0) = \frac{(n-\nu)! \nu!}{n!} a_{n-\nu},$$

and consequently

$$F_\nu(0) = \frac{\nu!(\nu-1)!(n-\nu)!(n-\nu-1)!}{n!^2} \left\{ \nu(n-\nu)a_{n-\nu}^2 - (\nu+1)(n-\nu+1)a_{n-\nu-1}a_{n-\nu+1} \right\}.$$

Furthermore the row of the $f_\nu(x)$ has only permanences for $x = +\infty$ and only variations for $x = -\infty$. Put

$$A_\nu = \nu(n-\nu)a_{n-\nu}^2 - (\nu+1)(n-\nu+1)a_{n-\nu-1}a_{n-\nu+1}, \quad A_0 = 1, \quad A_n = 1. \quad (3)$$

Then A_ν agrees, up to a positive factor, with $F_{n-\nu}(0)$, and the double row (7) of § 100, at $x = 0$ and written in reverse order, has the same signs as

$$A_0, \quad A_1, \quad A_2, \dots, \quad A_n. \quad (4)$$

In passing to $x = +\infty$ all variations in the row $f(x), f_1(x), \dots, f_n(x)$ are lost, and in passing to $x = -\infty$ all permanences are lost; hence in the double row of the f_ν, F_ν all VP ’s are lost for $+\infty$ and all PP ’s for $-\infty$. By § 100, II and III, we obtain:

V. The number of positive roots of $f(x)$ is equal to, or smaller by an even number than, the number of variation–permanences in the double row (4).

VI. The number of negative roots of $f(x)$ is equal to, or smaller by an even number than, the number of permanence–permanences in the double row (4).

The sum of the numbers of VP and PP is the number of permanences in the simple row

$$A_0, \quad A_1, \quad A_2, \dots, \quad A_n. \quad (5)$$

Thus V and VI combine into:

VII. The number of all real roots of $f(x)$ is equal to, or smaller by an even number than, the number of permanences in the row (5).

The total number of variations and permanences in (5) is n , the total number of real and imaginary roots together, provided no member of (5) vanishes. Since the first and last members have the same sign, the number of variations in (5) is even. Thus one may also add:

VIII. The number of imaginary roots of $f(x)$ is at least equal to the number of sign variations in the row (5).

In this form of the last two theorems it is immaterial whether some coefficients $a_1, a_2, \dots, a_n$ vanish, provided none of the A_ν is zero.

For the cubic

$$a_0x^3 + a_1x^2 + a_2x + a_3 = 0$$

one has

$$A_1 = 2(a_1^2 - 3a_0a_2), \quad A_2 = 2(a_2^2 - 3a_1a_3).$$

In the row 1, $A_1, A_2, 1$ there are two variations when one or both of A_1, A_2 is negative; in these cases the equation has two imaginary roots. If A_1 and A_2 are positive, the decision is not obtained from this theorem alone; one must then also consider

$$B = 9a_0a_3 - a_1a_2$$

and the sign of

$$D = A_1A_2 - B^2$$

(§ 62).

For a biquadratic equation one has the three expressions

$$A_1 = 3a_1^2 - 8a_0a_2, \quad A_2 = 4a_2^2 - 9a_1a_3, \quad A_3 = 3a_3^2 - 8a_2a_4.$$

If their signs are, in order, $+, -, +$, then there are four imaginary roots. If all three are positive, four, two, or no imaginary roots may occur; in all other cases there are two or four imaginary roots.

§ 102. Jacobi's Criterion

Jacobi derived from Descartes' rule a criterion for estimating the number of roots that lie between two limits α, β .⁴ If

$$f(x) = 0 \tag{1}$$

is the equation of degree n whose roots between α and β are to be counted, Jacobi puts

$$x = \frac{\alpha + \beta y}{1 + y} \tag{2}$$

and forms

$$(1 + y)^n f\left(\frac{\alpha + \beta y}{1 + y}\right) = \varphi(y) = b_0y^n + b_1y^{n-1} + \dots + b_n. \tag{3}$$

Here $b_0, b_1, \dots, b_n$ are integral homogeneous functions of degree n in α and β , and linear functions of the coefficients of $f(x)$. As x increases from α to β , the variable y also increases, from 0 to ∞ . Therefore $f(x)$ has precisely as many roots between α and β as $\varphi(y)$ has positive roots. Descartes' theorem gives:

IX. The number of roots of $f(x)$ between α and β is equal to, or smaller by an even number than, the number of sign variations in the row

$$b_0, \quad b_1, \quad b_2, \dots, \quad b_n.$$

⁴Jacobi, *Observatiunculæ ad theoriam æquationum algebraicarum pertinentes*. Crelle's Journal, vol. 13. Collected Works, vol. III.

Thus, whereas Budan's theorem requires the variations in two rows of values to be counted and compared, Jacobi's criterion requires only the counting of variations in one row. Newton's rule can be transformed in the same way. If

$$B_\nu = \nu(n - \nu)b_\nu^2 - (\nu + 1)(n - \nu + 1)b_{\nu-1}b_{\nu+1},$$

then one obtains:

X. The number of roots of $f(x)$ between α and β is equal to, or smaller by an even number than, the number of variation–permanences in the double row

$$b_0, b_1, b_2, \dots, b_n; \quad B_0, B_1, B_2, \dots, B_n.$$

§ 103. Geometrical Comparison of the Different Criteria

In order to compare the reach of the different criteria with one another, Klein used a geometrical interpretation, which we now discuss. If one does not wish to operate in spaces of more dimensions and thereby lose geometrical intuition, one must restrict oneself to the first cases. We begin with quadratic equations

$$x^2 + ax + b = 0. \quad (1)$$

Regard a and b as rectangular coordinates in a plane. Then every point of the plane is the image of one and only one quadratic equation, and conversely every quadratic equation with real coefficients has a point of the plane as its image.

The equation

$$a^2 - 4b = 0 \quad (2)$$

represents a parabola. On it lie the images of equations with equal roots. Points inside the parabola, that is, in its hollow, represent equations with imaginary roots; the outside points represent equations with real roots.

Following § 89, form the Sturm chain

$$x^2 + ax + b, \quad 2x + a, \quad a^2 - 4b. \quad (3)$$

For a fixed x , the equation

$$x^2 + ax + b = 0$$

is the tangent to the parabola at the point

$$a = -2x, \quad b = x^2.$$

Moreover $x^2 + ax + b$ is positive in the part of the plane that contains the parabola. This tangent and the parabola divide the plane into four regions, in which the row (3) has 0, 1, 2 variations as indicated in Fig. 16.

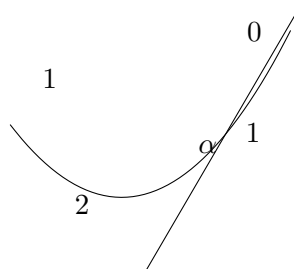


Fig. 16.

If we put $x = \alpha$ and $x = \beta$, we get two such tangents and a division of the plane into regions representing equations that have no, one, or two roots between α and β . In Fig. 17 these four regions are marked by the corresponding numerals. It is the distinction between points from which no tangent, one tangent, or two tangents can be drawn to the parabola with point of contact between α and β . Thus Fig. 17 gives the exact and complete distinction of the cases.

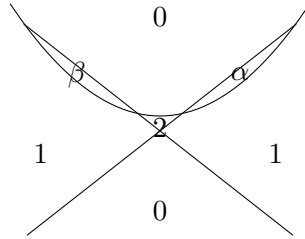


Fig. 17.

Compare this with Budan's theorem. We have to examine the three functions $f(x)$, $f'(x)$, $f''(x)$, hence

$$x^2 + ax + b, \quad 2x + a, \quad 1, \quad (4)$$

with respect to sign variations. The two straight lines

$$x^2 + ax + b = 0, \quad 2x + a = 0$$

divide the plane into three regions in which the row (4) has the variations indicated in Fig. 18.

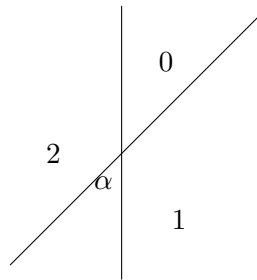


Fig. 18.

Taking again two values $x = \alpha$ and $x = \beta$, we obtain six regions, in which the loss of variations is shown in Fig. 19. In the regions marked 0 and 1, this loss gives the correct number of roots. In the region marked 2, however, the decision between two roots and no root is not made.

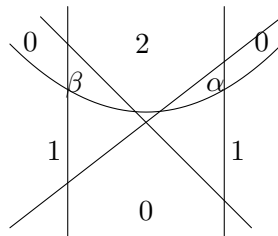


Fig. 19.

Now consider Jacobi's criterion (§ 102). To estimate the number of roots between α and β , the equation (1) is first transformed into

$$(\alpha + \beta y)^2 + a(\alpha + \beta y)(1 + y) + b(1 + y)^2 = 0$$

or

$$(\beta^2 + a\beta + b)y^2 + \{2\alpha\beta + a(\alpha + \beta) + 2b\}y + (\alpha^2 + a\alpha + b) = 0.$$

Thus one must count the variations in

$$\beta^2 + a\beta + b, \quad 2\alpha\beta + a(\alpha + \beta) + 2b, \quad \alpha^2 + a\alpha + b.$$

The first and third of these expressions, when set equal to zero, give the two tangents already considered. The middle expression

$$2\alpha\beta + a(\alpha + \beta) + 2b$$

vanishes at $a = -2\alpha$, $b = \alpha^2$, and at $a = -2\beta$, $b = \beta^2$; hence it represents the line joining the two points α, β , and it is negative on the side containing the point of intersection of the two tangents. At that intersection point it is itself equal to $-(\beta - \alpha)^2$.

We therefore obtain five regions with zero, one, or two variations, as shown in Fig. 20.

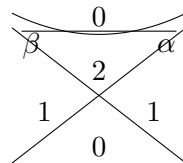


Fig. 20.

One sees that this figure confines the indeterminacy to a much smaller region than the previous one. From this standpoint it does not appear justified that Jacobi gives such decided preference to the Budan–Fourier criterion over his own.⁵

§ 104. Determination of an Upper Bound for the Roots

In the third section we already used the fact that one can always find a positive number greater than the absolute values of all roots of a given equation. What is now required is a determination of such an upper bound that is as simple and at the same time as sharp as possible.

We first consider only real roots and seek a positive number greater than the largest positive root of a given equation $f(x) = 0$. Budan's theorem gives such a determination. Assume the coefficient of the highest power of x in $f(x)$ equal to 1 (or at least positive). The positive number α can always be taken so large that

$$f(\alpha), \quad f'(\alpha), \quad f''(\alpha), \dots, \quad f^{(n)}(\alpha) \tag{1}$$

are all positive. Then in the row

$$f(x), \quad f'(x), \quad f''(x), \dots, \quad f^{(n)}(x)$$

from $x = \alpha$ to $x = \infty$ no variation can be lost, and therefore by Theorem I no root of $f(x)$ can lie between α and ∞ .

For a lower bound for the negative roots take a positive number β such that

$$(-1)^n f(-\beta), \quad (-1)^{n-1} f'(-\beta), \dots, \quad f^{(n)}(-\beta) \tag{2}$$

are all positive. Then $f(x) = 0$ has certainly no negative root below $-\beta$.

This determination of bounds already goes back to Newton. Another upper bound, simpler in calculation, was given by Laguerre.⁶ Instead of the derivatives he uses the functions

⁵Compare F. Klein, *Geometrisches zur Abzählung der reellen Wurzeln einer algebraischen Gleichung*, in Dyck's catalogue of mathematical models (Munich 1892).

⁶Laguerre, *Nouvelles Annales de math.*, second series, vol. XIX, 1880; *Journal de math.*, third series, vol. IX, 1883.

$f_0(x), f_1(x), f_2(x), \dots, f_n(x)$, which have already served us several times, especially in the Tschirnhausen transformation.

These functions were defined in § 4 by

$$\begin{aligned} f_0(x) &= 1, \\ f_1(x) &= x + a_1, \\ f_2(x) &= x^2 + a_1x + a_2, \\ &\vdots \\ f_{n-1}(x) &= x^{n-1} + a_1x^{n-2} + a_2x^{n-3} + \dots + a_{n-1}, \\ f(x) = f_n(x) &= x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n, \end{aligned}$$

with the recursion

$$f_\nu(x) - xf_{\nu-1}(x) = a_\nu.$$

As shown there,

$$f(x) = (x - \alpha)\{x^{n-1}f_0(\alpha) + x^{n-2}f_1(\alpha) + \dots + f_{n-1}(\alpha)\} + f(\alpha). \quad (3)$$

This formula immediately shows that if α makes

$$f_0(\alpha), \quad f_1(\alpha), \quad f_2(\alpha), \dots, \quad f_{n-1}(\alpha), \quad f(\alpha) \quad (4)$$

positive, then no positive x greater than α can make $f(x)$ vanish, since the right side of (3) then consists only of positive terms. Thus such an α is an upper bound for the positive roots of $f(x)$.

Likewise one obtains a lower bound $-\beta$ for the negative roots if $\beta > 0$ is so chosen that

$$f_0(-\beta), \quad -f_1(-\beta), \quad f_2(-\beta), \dots, \quad (-1)^n f(-\beta) \quad (5)$$

are all positive. This determination is easier in calculation, but it can give less sharp bounds than Newton's. Indeed, differentiating (3) repeatedly with respect to x and then putting $x = \alpha$, under the assumption that all quantities in (4) are positive, gives expressions for $f'(\alpha), f''(\alpha), \dots$ consisting of positive terms. Hence if (4) is positive, then the derivatives (1) are necessarily all positive; the converse is not necessary.

To find an upper bound for the absolute value of the imaginary roots, valid also for complex coefficients, consider

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n,$$

where $a_1, a_2, \dots, a_n$ may be real or imaginary. Denote absolute value by $|a|$. If $|x| > 1$, then

$$\left| a_1 + \frac{a_2}{x} + \dots + \frac{a_n}{x^{n-1}} \right| < |a_1| + |a_2| + \dots + |a_n| = \alpha.$$

Thus, if $|x|$ is greater than both 1 and α , the expression

$$x^{1-n}f(x) = x + a_1 + \frac{a_2}{x} + \dots + \frac{a_n}{x^{n-1}}$$

cannot vanish, because the absolute value of x is greater than the absolute value of the sum of all other terms on the right. Hence, according as α is greater or less than 1, if one draws about the origin a circle of radius α or of radius 1, no root of $f(x)$ lies outside this circle.

§ 105. Estimation of the Imaginary Roots

In § 96 we saw how the characteristic of a certain system of curves determines the exact number of complex roots of an equation that lie inside a closed curve. This test too can be simplified when one seeks not the exact number of roots but only an upper bound for it, which in many cases is already enough for exact determination.

Let, as in § 96,

$$F(z) = \varphi(x, y) + i\psi(x, y) \quad (1)$$

be a real or complex function of the complex argument $z = x + iy$, with φ and ψ real functions of x and y , and let a closed curve f be drawn. We have:

XI. If the curve f is divided by the curve φ into $2m$ arcs in which φ is alternately positive and negative, then the number of roots of $F(z)$ inside f is at most m . The same holds if f is divided by ψ into $2m$ such arcs.

This is an immediate consequence of the theorem on characteristics II, § 96. It gives

$$k = \frac{1}{2}\{E(f; \varphi, \psi) - A(f; \varphi, \psi)\}.$$

If f is cut by the φ -curve in $2m$ points, then

$$m = \frac{1}{2}\{E(f; \varphi, \psi) + A(f; \varphi, \psi)\},$$

and hence

$$k = m - A(f; \varphi, \psi).$$

Here k is, by the cited theorem, the number of roots of F inside f , and $A(f; \varphi, \psi)$ is certainly non-negative. The same argument applies to

$$k = \frac{1}{2}\{E(f; \psi, \varphi) - A(f; \psi, \varphi)\}.$$

§ 106. Rolle's Theorem

Suppose that $f(x) = 0$ has two real roots α and β , the first of multiplicity a and the second of multiplicity b . Then

$$f(x) = (x - \alpha)^a(x - \beta)^b f_1(x). \quad (1)$$

If $\alpha < \beta$ and no root of $f(x) = 0$ lies between α and β , then $f(x)$ and $f_1(x)$ have constant sign throughout this interval. Differentiating (1), one obtains

$$\frac{(x - \alpha)(x - \beta)f'(x)}{f(x)} = a(x - \beta) + b(x - \alpha) + (x - \alpha)(x - \beta)\frac{f_1'(x)}{f_1(x)}. \quad (2)$$

The right side is negative for $x = \alpha$, namely $a(\alpha - \beta)$, and positive for $x = \beta$, namely $b(\beta - \alpha)$. As x passes from α to β it must therefore pass through zero once, or more generally an odd number of times. On the left side of (2), $(x - \alpha)(x - \beta)$ and $f(x)$ have constant sign, and consequently $f'(x)$ must vanish an odd number of times. This gives Rolle's theorem:

XII. If α and β are two consecutive roots of $f(x) = 0$, then between α and β , endpoints excluded, there lie roots of $f'(x) = 0$ in odd number, hence at least one.

Important consequences follow for the case in which $f(x) = 0$ has only real roots. Let these be ordered by size as

$$\alpha, \beta, \gamma, \dots,$$

and suppose that α is of multiplicity a , β of multiplicity b , γ of multiplicity c , and so on. Then $f'(x) = 0$ has at α a root of multiplicity $a - 1$, at β a root of multiplicity $b - 1$, at γ a root of multiplicity $c - 1$, and between α and β , between β and γ , and so forth, at least one real root. The total number so obtained is

$$a + b + c + \dots - 1 = n - 1,$$

and because $f'(x)$ has degree $n - 1$, this minimum number is not exceeded. Thus:

XIII. If $f(x) = 0$ has only real roots, then $f'(x) = 0$ also has only real roots; those which do not coincide with multiple roots of $f(x)$ are simple and separate the roots of $f(x)$.

Repeated application gives the general form:

XIV. If $f(x) = 0$ has only real roots, then $f'(x), f''(x), f'''(x), \dots$ also have only real roots. If one of these functions $f^{(\nu)}(x)$ has a root α of multiplicity a , and $a > 1$, then α is a root of $f(x)$ of multiplicity $a + \nu$.

These theorems are essentially generalized by applying a linear transformation.⁷ The matter is most easily handled in homogeneous notation: let $f(x, y)$ be a homogeneous function of degree n . For an arbitrary value $\xi : \eta$ form the polars (§ 60)

$$\begin{aligned} P_1(x, \xi) &= \frac{1}{n} \{ \xi f'(x) + \eta f'(y) \}, \\ P_2(x, \xi) &= \frac{1}{n(n-1)} \{ \xi^2 f''(x, x) + 2\xi\eta f''(x, y) + \eta^2 f''(y, y) \}, \\ P_3(x, \xi) &= \frac{1}{n(n-1)(n-2)} \{ \xi^3 f'''(x, x, x) + 3\xi^2\eta f'''(x, x, y) \\ &\quad + 3\xi\eta^2 f'''(x, y, y) + \eta^3 f'''(y, y, y) \}. \end{aligned} \tag{3}$$

These functions are unchanged if ξ, η and x, y are transformed simultaneously by a linear transformation, and if f and its derivatives are replaced by the corresponding transformed functions. The reality of the roots is not changed by a real linear transformation. If ξ, η are real, the transformation may be chosen so that $\xi' = 1, \eta' = 0$; after also putting $y = 1$, the polars become the derivatives of $f(x)$ with respect to x , apart from numerical factors.

Thus in XIII and XIV the derivatives $f'(x), f''(x), \dots$ may be replaced by the polars $P_1(x, \xi), P_2(x, \xi), \dots$ for arbitrary real ξ, η .

Apply this to the equation obtained by setting the $(n - 2)$ nd polar equal to zero:

$$x^2 f''(\xi, \xi) + 2xy f''(\xi, \eta) + y^2 f''(\eta, \eta) = 0. \tag{4}$$

If, as assumed, $f(x, y) = 0$ has only real roots, then this quadratic equation must have real roots. Hence the Hessian determinant

$$f''(\xi, \xi) f''(\eta, \eta) - f''(\xi, \eta)^2 \tag{5}$$

must be negative for all real ξ, η . It can vanish only if (4) has equal roots, and by XIV this is possible only when $f(x, y)$ is the n th power of a linear function. In that exceptional case the determinant (5) vanishes identically. Apart from this case, therefore, for an equation with only real

⁷Laguerre, *Nouvelles Annales de Mathématiques*, second series, vol. 19, 1880.

roots, the equation obtained by setting the Hessian determinant equal to zero has only imaginary roots.

Returning to inhomogeneous notation, Euler's theorem gives

$$\begin{aligned} xf'(x) + yf'(y) &= nf(x), \\ xf''(x, x) + yf''(x, y) &= (n-1)f'(x), \\ xf''(y, x) + yf''(y, y) &= (n-1)f'(y). \end{aligned} \quad (6)$$

Eliminating $f'(y)$, $f''(x, y)$, $f''(y, y)$, and then putting $y = 1$, one obtains

$$\begin{aligned} f'(y) &= nf(x) - xf'(x), \\ f''(x, y) &= (n-1)f'(x) - xf''(x), \\ f''(y, y) &= n(n-1)f(x) - 2(n-1)xf'(x) + x^2f''(x). \end{aligned} \quad (7)$$

Thus, if

$$(n-1)H(x, y) = f''(x, x)f''(y, y) - f''(x, y)^2, \quad (8)$$

then

$$H(x, y) = H(x) = nf(x)f''(x) - (n-1)f'(x)^2. \quad (9)$$

The degree of $H(x)$ in x is at most $2n-4$.

§ 107. Laguerre's Theorems for Equations with Only Real Roots

Laguerre made an application of the preceding theorems to equations with only real roots. Let $f(x) = 0$ be an equation of degree n with n distinct real roots

$$\alpha, \alpha_1, \alpha_2, \dots$$

For a variable x we have the familiar formula (§ 14, (11))

$$\frac{f'(x)}{f(x)} = \sum \frac{1}{x - \alpha}, \quad (1)$$

where the sum runs over the different roots. Differentiating once more gives

$$\sum \frac{1}{(x - \alpha)^2} = \frac{f'(x)^2 - f(x)f''(x)}{f(x)^2}. \quad (2)$$

Introducing H from § 106, (9), this becomes

$$\sum \frac{1}{(x - \alpha)^2} = \frac{nf'(x)^2 - H(x)}{nf(x)^2}. \quad (3)$$

A linear transformation in which the value $x = \infty$ corresponds to an arbitrary other real value of the new variable gives a more general formula. Introduce homogeneous variables by replacing x by $x : y$ and α by $\alpha : \beta$. Applying the linear transformation

$$\begin{aligned} x &= ax' + by', & \alpha &= a\alpha' + b\beta', \\ y &= cx' + dy', & \beta &= c\alpha' + d\beta', \end{aligned} \quad (5)$$

or, inversely,

$$rx' = dx - by, \quad ry' = -cx + ay, \quad r = ad - bc,$$

and using the covariance of H , one obtains, after dropping primes and writing again f ,

$$\sum \left(\frac{\xi\beta - \eta\alpha}{x\beta - y\alpha} \right)^2 = \frac{[\xi f'(x) + \eta f'(y)]^2 - H(x, y)}{nf(x, y)^2}. \quad (8)$$

In this form both sides are unchanged by any simultaneous linear transformation of the three pairs $x, y; \xi, \eta; \alpha, \beta$.

Formula (8) is an identity for every form $f(x, y)$ and for all values of the variables. Now assume that $f(x, y)$ has only real roots, so that the α, β are real, and let also ξ, η and x, y be real. Both sides are then essentially positive. Denote their common value by P and determine $X : Y$ from

$$\frac{(X\eta - Y\xi)^2}{(Xy - Yx)^2} = P,$$

or

$$\Omega = (Xy - Yx)^2 P - (X\eta - Y\xi)^2 = 0. \quad (9)$$

This quadratic equation has two real roots.

If $X = x, Y = y$, then Ω is negative. If $X = \alpha, Y = \beta$ is any root of $f = 0$, then the corresponding quotient is one term of the sum P and is therefore smaller than P ; hence Ω is positive. Ordering the points $\alpha : \beta$ and $x : y$ cyclically on the circle and writing the variable value $X : Y$ as a moving point, Ω changes sign between x and the nearest roots on either side. If X_1, X_2 are the roots of (9), their order is

$$\alpha_1, \quad X_1, \quad x, \quad X_2, \quad \alpha_2. \quad (10)$$

Thus (X_1, X_2) is an interval containing x but no root of $f = 0$; X_1 and X_2 lie closer to the nearest roots α_1, α_2 than the original value x .

The point ξ may have any position. The points X_1, X_2 depend on it, but they always lie in (α_1, x) and (x, α_2) respectively.

The question is how to choose ξ so that X_1 is as close as possible to α_1 , or X_2 as close as possible to α_2 . Equivalently, with X fixed, (9) is a quadratic equation for ξ ; admissible boundary positions occur when its discriminant vanishes. The coefficients of $\xi^2, 2\xi\eta, \eta^2$ in (9) are

$$\begin{aligned} & \frac{f'(x)^2 - y^2 H}{nf^2} (Xy - Yx)^2 - Y^2, \\ & \frac{f'(x)f'(y) + xyH}{nf^2} (Xy - Yx)^2 + XY, \\ & \frac{f'(y)^2 - x^2 H}{nf^2} (Xy - Yx)^2 - X^2. \end{aligned}$$

Using $xf'(x) + yf'(y) = nf$, the discriminant condition gives

$$\frac{(Xy - Yx)^2}{nf^2} \left\{ (n-1)H(Xy - Yx)^2 + [Xf'(x) + Yf'(y)]^2 \right\} = 0.$$

Therefore the limiting values of $X : Y$ satisfy

$$[Xf'(x) + Yf'(y)]^2 + (n-1)(Xy - Yx)^2 H = 0. \quad (11)$$

Since H is negative, this quadratic equation has two real roots, determined by

$$Xf'(x) + Yf'(y) = \pm (Xy - Yx) \sqrt{(1-n)H}. \quad (12)$$

Several special results follow on leaving the homogeneous notation. First take $y = 0$, so that $x : y$ is infinite, and put $Y = 1$. Then two values X_1, X_2 are obtained between which all roots of $f(x)$ are contained. If

$$f(x, y) = a_0 x^n + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + \cdots,$$

then

$$H = \{2na_0 a_2 - (n-1)a_1^2\} x^{2n-4}, \quad (13)$$

and these two values are obtained from

$$na_0X = -a_1 \pm \sqrt{(n-1)^2a_1^2 - 2n(n-1)a_0a_2}. \quad (14)$$

If, however, $x : y$ is left arbitrary and we return to the inhomogeneous form by putting $y = 1$, $Y = 1$, and using (7) and (9) of § 106, then (12) gives

$$x - X = \frac{nf(x)}{f'(x) \pm \sqrt{(n-1)^2f'(x)^2 - n(n-1)f(x)f''(x)}}. \quad (15)$$

These values have the following meaning. If x lies between two roots α_1, α_2 of $f(x)$, then of the two values of X one lies nearer α_1 and the other nearer α_2 , both still inside (α_1, α_2) . If x is greater than the greatest root or smaller than the least root, then all roots lie between the two values of X , while x lies outside. If x is close to a root, one of the two values of X gives a still more accurate value of that root.

Just as X_1, X_2 determine an interval containing no root, one can also determine an interval in which roots certainly lie. Let $f(x, y)$ again be a form of degree n with only real roots, and let $x : y$, $\xi : \eta$, $\xi' : \eta'$ be three real values, two arbitrary, the third related by

$$[\xi f'(x) + \eta f'(y)][\xi' f'(x) + \eta' f'(y)] - H(\xi y - \eta x)(\xi' y - \eta' x) = 0. \quad (16)$$

This relation is invariant under simultaneous linear transformations. Reducing it to a special position gives

$$\xi' = \frac{\sum(\alpha^2)}{\sum(\alpha)}$$

and equivalently

$$\sum \alpha(\alpha - \xi') = 0.$$

Since all α are real, the products $\alpha(\alpha - \xi')$ cannot all be positive nor all negative. Thus some roots lie between 0 and ξ' , and some outside.

By transforming back, one obtains the general conclusion: on the circle carrying the points α, x, ξ, ξ' , with ξ, ξ' connected by (16), each of the two arcs into which the circle is divided by ξ and ξ' contains at least one root α . The point x and one of ξ, ξ' are arbitrary; the third is determined by (16).

In particular, let ξ' coincide with one of the two points X determined by (12), say

$$Xf'(x) + Yf'(y) + (Xy - Yx)\sqrt{(1-n)H} = 0. \quad (17)$$

Putting $\xi', \eta' = X, Y$ in (16) and cancelling $Xy - Yx$ gives

$$(n-1)[\xi f'(x) + \eta f'(y)] + (\xi y - \eta x)\sqrt{(1-n)H} = 0, \quad (18)$$

which determines ξ uniquely.

In inhomogeneous form, (17) and (18) give

$$x - X = \frac{nf(x)}{f'(x) + (n-1)\sqrt{f'(x)^2 - \frac{n}{n-1}f(x)f''(x)}}, \quad (19)$$

$$x - \xi = \frac{nf(x)}{f'(x) + \sqrt{f'(x)^2 - \frac{n}{n-1}f(x)f''(x)}}, \quad (20)$$

where the square roots may be taken with either sign.

The position of the points is then described as follows. There is at least one root α such that

$$\xi, \quad \alpha, \quad X, \quad x \quad (21)$$

occur in this order, and no root lies between x and X . One may choose the root so that no further root lies between α and X . Between α and ξ , however, other roots may occur.

Let now x approach the root α . Then X approaches the same root, and (20) shows that ξ approaches it as well. Hence, when x is sufficiently close to a root, this nearest root lies in the interval (ξ, X) , which contains no second root, and the interval closes more and more around α .

As an example take the spherical functions, which we earlier proved (§ 87) to have only real roots, all between -1 and $+1$. The function

$$P_n(x) = \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{1 \cdot 2 \cdot \dots \cdot n} \left(x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \dots \right)$$

has

$$P_n(1) = 1, \quad P'_n(1) = \frac{n(n+1)}{2}, \quad P''_n(1) = \frac{n(n-1)(n+1)(n+2)}{8}.$$

This is easily proved by induction, using the recursion

$$(1-x^2)P'_n(x) + nxP_n(x) - nP_{n-1}(x) = 0.$$

Putting $x = 1$ in (19) and (20) gives two bounds enclosing the root nearest to 1:

$$X = 1 - \frac{2}{n+1 + (n-1)\sqrt{\frac{n(n+1)}{2}}},$$

$$\xi = 1 - \frac{2}{n+1 + \sqrt{\frac{n(n+1)}{2}}}.$$

For $n = 7$, for example,

$$X = 0.94967\dots, \quad \xi = 0.84952\dots$$

The more exact values of the two roots nearest to 1, listed for comparison from Gauss' computation (Gauss, Works, vol. III, p. 195), are

$$0.94911\dots; \quad 0.74153\dots$$

Thus here only one root lies between X and ξ , and X exceeds the value of that root only in the fourth decimal.